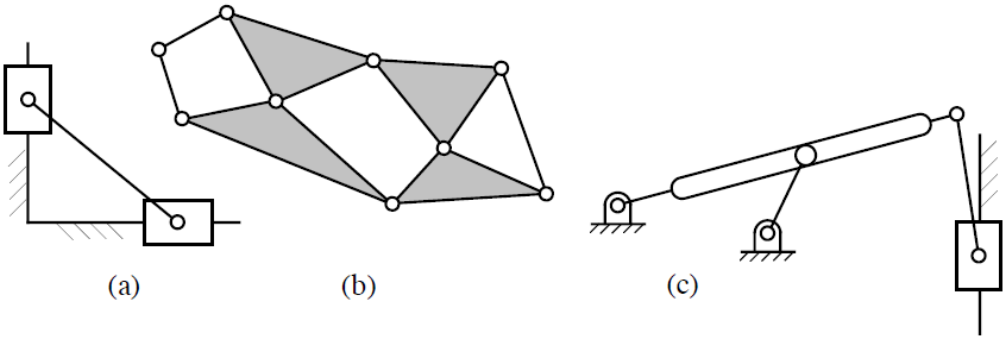
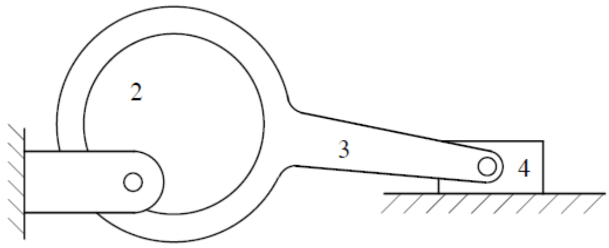
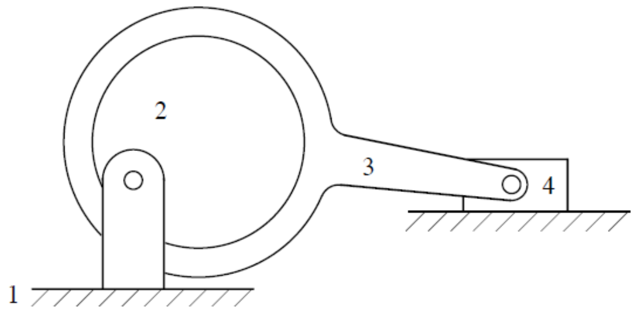
What is the number of members, number of joints, and mobility of each of the planar linkages shown below?

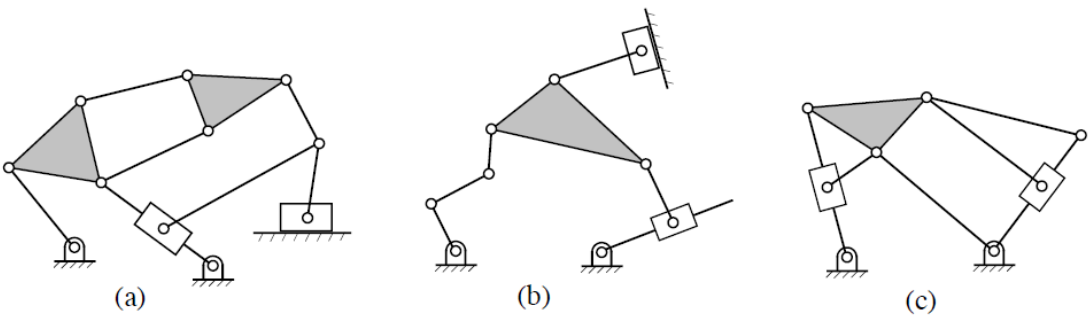


Determine the mobility and the number of idle degrees of freedom associated with the mechanism. Show the equations used and identify any assumptions made when determining your answers.

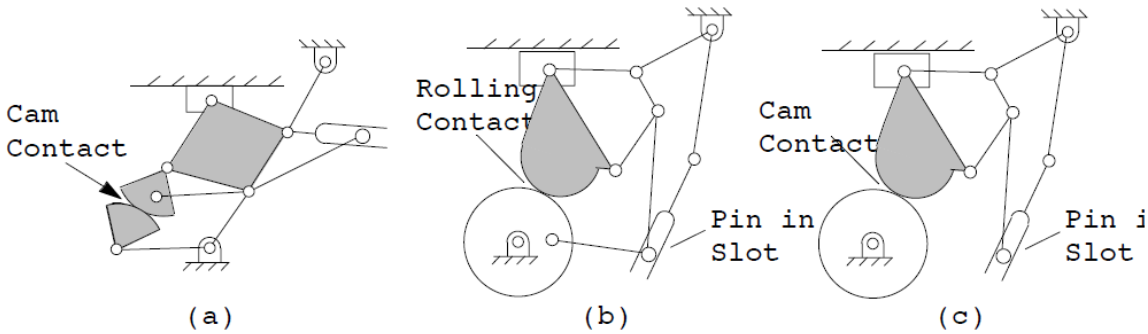




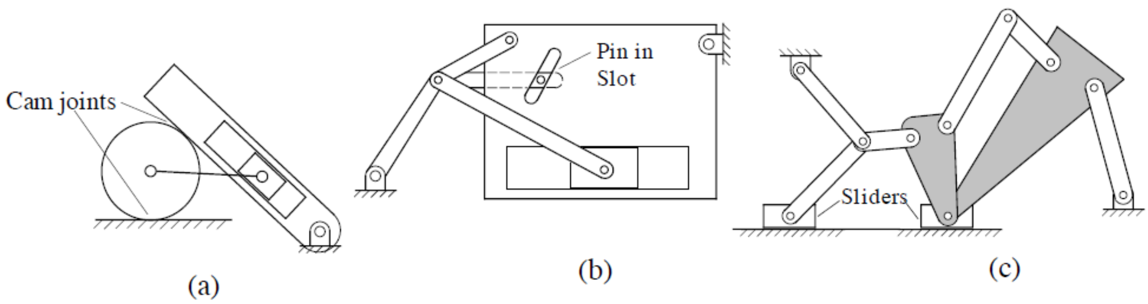
Determine the mobility and the number of idle degrees of freedom of each of the planar linkages shown below. Show the equations used to determine your answers.



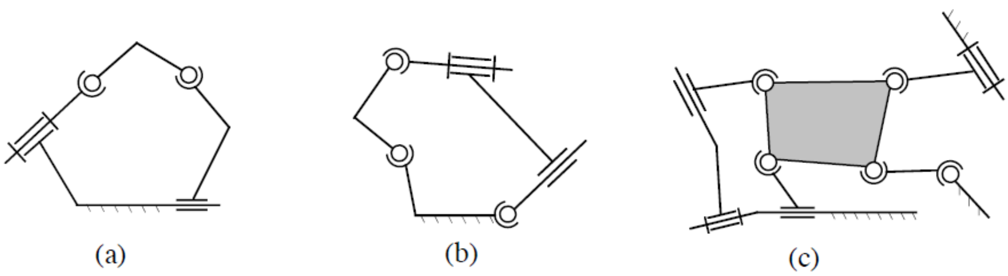
Determine the mobility and the number of idle degrees of freedom for each of the mechanisms shown. Show the equations used and identify any assumptions made when determining your answers.



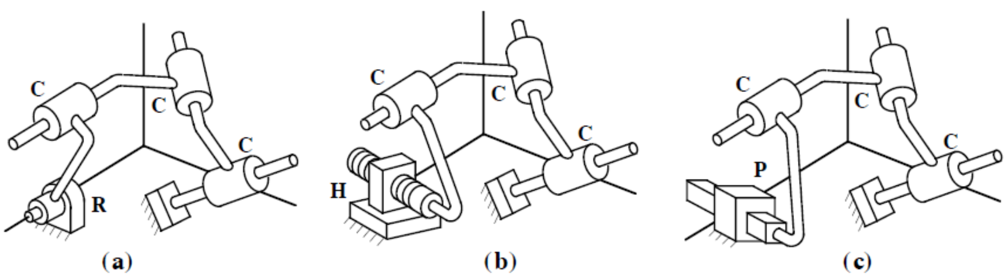
Determine the mobility and the number of idle degrees of freedom associated with each mechanism. Show the equations used to determine your answers.



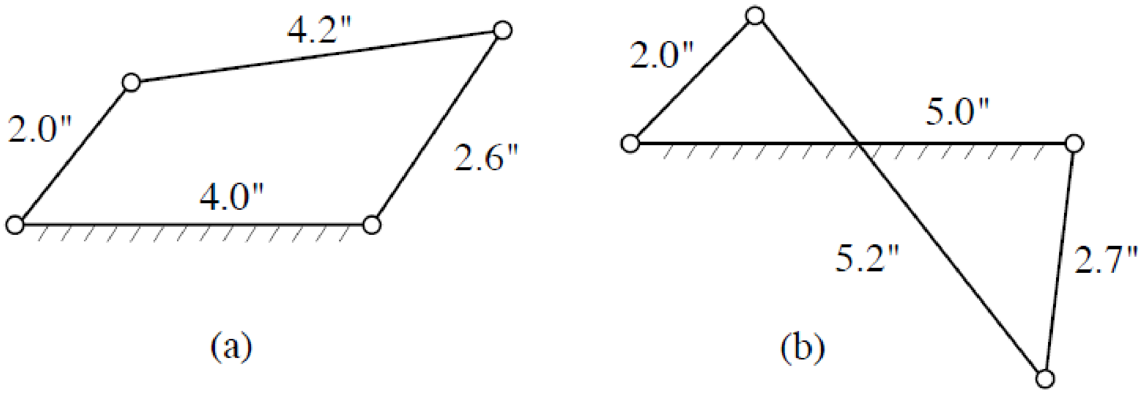
Determine the mobility and the number of idle degrees of freedom of the spatial linkages shown below. Show the equations used to determine your answers.



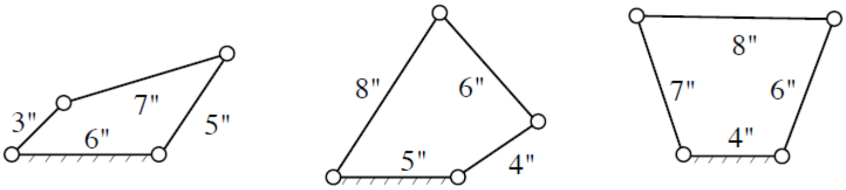
Determine the mobility and the number of idle degrees of freedom associated with each mechanism.[3] Show the equations used to determine your answers.



If these linkages are being driven by link 2 find the angles measured in anticlockwise sense with the horizontal that are the two extreme positions for link 4. Use simple rough sketches and trigonometry rather than accurate drawings. As per Grashof criterion what kind of linkages are these ?

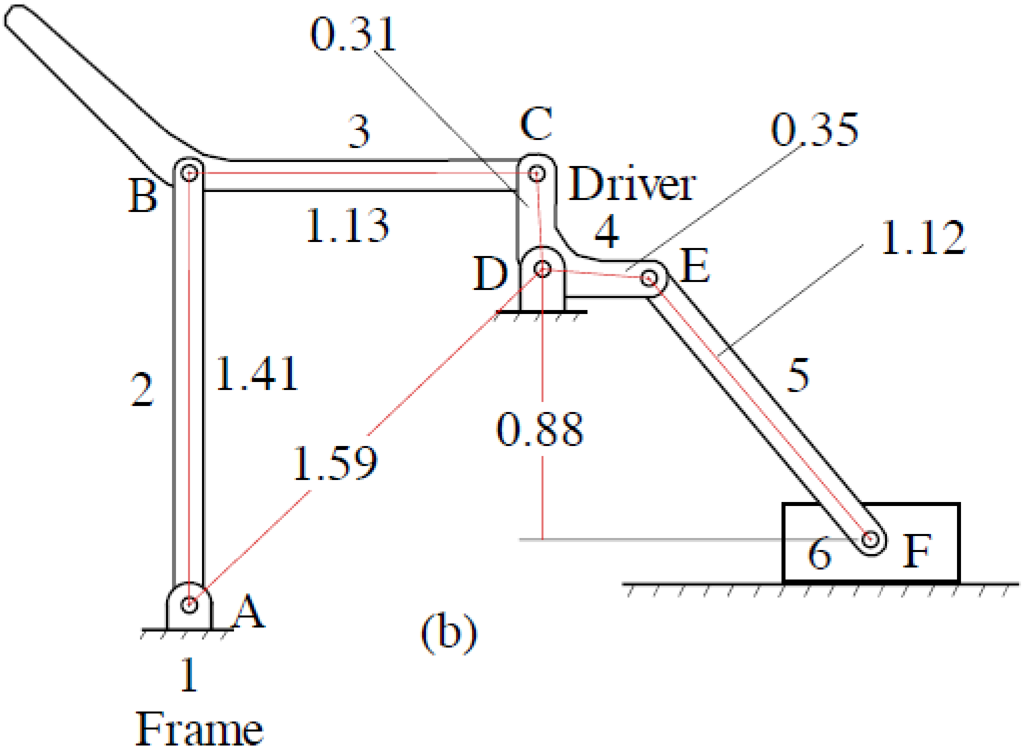


For the four-bar linkages below, indicate whether they are Grashof type 1 or 2 and whether they are crank-rocker, double-crank, or double-rocker mechanisms.



(a) (b) (c)

Also find out the angle(s) made by link 2 (the leftmost link) with the horizontal as measured in the anticlockwise sense in the dead center configuration(s) if such configurations exist. Try to solve the same problem when the links are in a crossed configuration. Do you get the same answer? Use simple rough sketches and trigonometry rather than accurate drawings.



If the link 2 is the input link find the two extreme positions of the slider F. Repeat the same exercise if link 4 is the input link.